

SUSHMA ACHARYA

Computer Engineering Undergraduate

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PROFILE

Computer Engineering undergraduate at Kathmandu University with hands-on experience across full-stack web development, and interface design. Built end-to-end products spanning web applications, NLP, real-time 3D visualization, and UI/UX reflecting both technical range and a consistent drive to turn ideas into working products. Seeking an internship opportunity where I can keep learning quickly and contribute from day one.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C++, C#

Frontend: React.js, Next.js, Tailwind CSS

Backend: Node.js, Express.js, FastAPI, RESTful APIs

AI / ML & Data: PyTorch, spaCy, Deep Learning (DenseNet121, ViT), NLP, PostgreSQL, MongoDB, MySQL

Tools & Platforms: Git, GitHub, Docker, Figma, Unity Engine, Three.js

PROJECTS

KaamLink: Job Portal Web App | *React, Node.js, Express.js, MongoDB, Redux Toolkit*

- Developed a full-stack job portal with separate job-seeker and recruiter workflows for browsing, posting, saving, and applying to jobs.

Medical Vision Assistant | *Next.js, TypeScript, FastAPI, PostgreSQL, PyTorch, Docker, DenseNet121, ViT*

- Designed and built an end-to-end AI platform for chest X-ray analysis, combining disease classification, GradCAM explainability, automated report generation, and a patient-history dashboard.
- Trained and integrated DenseNet121 and Vision Transformer (ViT) models for classification; containerized the full stack with Docker Compose for easy deployment.

SaralDoc: Nepali Legal Document Analyzer | *React, FastAPI, Python, spaCy/NLP, PostgreSQL*

- Built a full-stack tool that extracts clauses, obligations, and named entities (dates, amounts, parties) from Nepali and English legal documents.
- Implemented PDF/DOCX/TXT ingestion, a document-complexity score, and automatic summarization to make dense legal text easier to read.

Multimodal AI Assistant | *Next.js, TypeScript, Tailwind CSS, Gemini API*

- Built a multimodal AI chatbot supporting image uploads, Markdown rendering, and user authentication behind a clean, responsive interface.

Water Cycle Simulator | *React, Three.js*

- Created an interactive 3D educational simulator visualizing evaporation, condensation, precipitation, and collection with custom particle systems and multi-camera controls.

GROUP PROJECTS:

Rush Rally | *Unity, C#, ShaderLab*

- Collaborated on a team-built racing game, contributing to in-game assets, scene design, coin/collectible systems, and core gameplay mechanics.

GhumFir | *Next.js*

- Designed and implemented responsive frontend interfaces in TypeScript/React, integrating third-party location and travel APIs for dynamic itineraries and user interaction.

Heritage Damage Assessment | *React*

- Engineered the interactive frontend dashboard using React and connected RESTful API endpoints with a FastAPI backend to display automated heritage site damage classifications and temporal deterioration metrics

EDUCATION

Bachelor of Computer Engineering

Kathmandu University

2022 – 2027

Higher Secondary Education

St. Xavier's College, GPA: 3.68

2020 – 2022

ADDITIONAL

KU HackFest 2025: GarbageGO | *Unity, C#, AI/ML (ML-Agents)*

- Developed an AR gamified mobile application using Unity, C#, and Unity ML-Agents to detect real-world trash in real-time, spawning interactive gamified targets to incentivize proper waste disposal.